

WASP-12b

R-2431

Benchmark run · workflow W8-benchmark-deep · record deep-bench_WASP-12_141025 · created 2026-08-22T00:40:26+00:00

BENCHMARK: NON-DETECTION

Results

Mid-Transit Time (Tmid)	2456954.84955 +/- 0.0001 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.037 +/- 0.045
Transit depth (Rp/Rs)^2	0.14 +/- 0.33 [%]
Orbital Inclination (inc)	85.07 +/- 2.81
Airmass coefficient 1 (a1)	4.84 +/- 3.12
Airmass coefficient 2 (a2)	-1.11 +/- 0.37
Scatter in the residuals of the lightcurve fit is	65.56 %
Best Comparison Star	#3 - [71, 272]
Optimal Aperture	3.0
Optimal Annulus	6.5
Transit Duration (day)	0.1172 +/- 0.0065

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.037 ± 0.045 vs 0.1178 (deviation 1.8σ)
Duration fit vs published	0.1172 d vs 0.125 d (deviation 1.2σ)
Mid-time O–C	-1571.6 min
Depth SNR	0.8
Residual scatter	65.56 %

Light curve

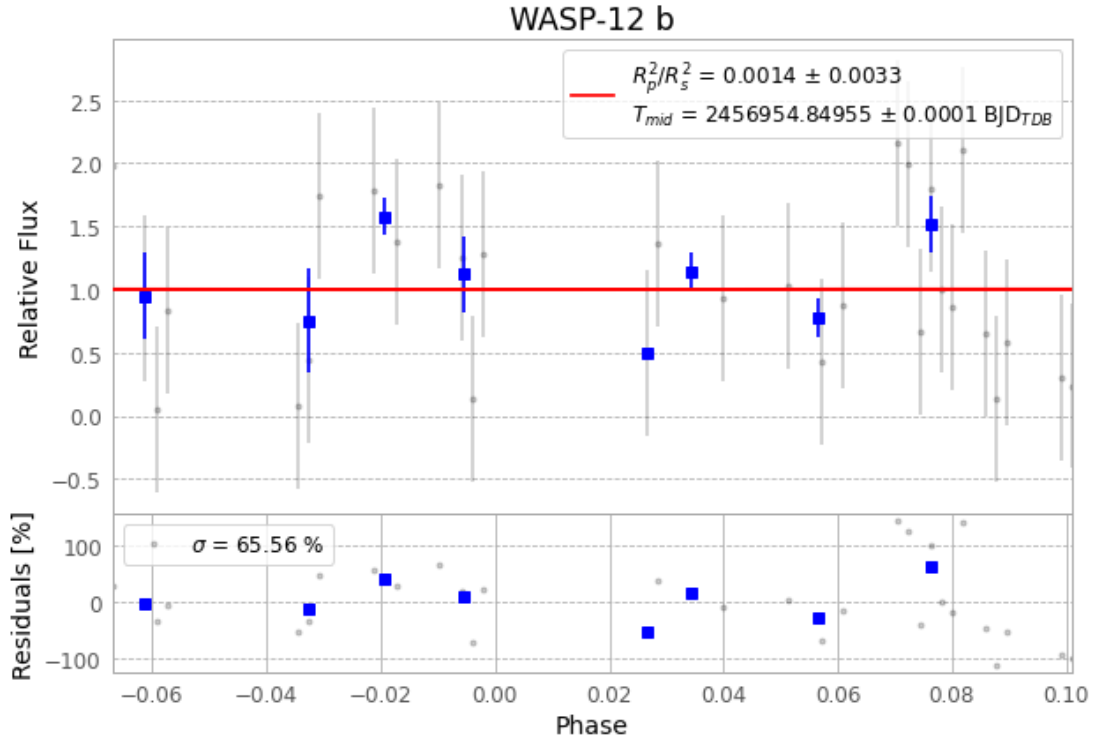


Figure 1 — FinalLightCurve_WASP-12 b_2014-10-23.png (SHA-256 34d3323020a1c10e...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [198, 323], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 e4e55b6e4b1c78ae...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit a4253ce

Data provenance

90 FITS frames (59 MB), WASP-12141024063312.FITS → WASP-12141024110012.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-12-141025.log (5bbdfa90e7dc...), FinalLightCurve_WASP-12 b_2014-10-23.png (34d3323020a1...), FinalLightCurve_WASP-12 b_2014-10-23.csv (e50efa7834d2...)

Compute resources

Wall time 113.0 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-22 00:40Z.